Short-horizon mean reversion in cryptocurrency markets: a kernel-constrained logit measurement of directional predictability

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Abstract

We measure out-of-sample directional (sign-of-return) predictability in cryptocurrency markets at 15-minute to 4-hour horizons, contrasting it against US equities, with a three-parameter autoregressive logit whose lag weights are tied to a smooth decaying kernel (algebraically the Glauber form of a kinetic Ising chain), used as a low-variance measurement device. One matched walk-forward protocol, with moving-block-bootstrap inference and false-discovery-rate control, is applied at population scale to 183 cryptocurrency pairs versus 187 US stocks/ETFs (a focal panel of fifteen instruments across six asset classes provides context; the FX, commodity, and bond readings are suggestive only). Fifteen-minute predictability is significant in 90% of crypto pairs versus 3% of stocks, with a dependence-aware class-mean AUC gap of +0.031 (95% CI $[0.027, 0.035]$, stable across bootstrap block lengths). The gap is not a session artifact, since restricting the entire stock universe to regular trading hours leaves it at AUC 0.498. Nor is it an artifact of the estimator, because a parameter-free multi-lag *sign* autocorrelation reproduces it (class gap -0.035, 95% CI $[-0.041, -0.029]$), as does the DFA Hurst exponent, even though the lag-1 *return* autocorrelation is $\approx$0; we show why the two disagree. The fitted coupling is mean-reverting in 98% of crypto pairs and near zero in equities, with FX intermediate. A cross-sectional regression shows the crypto edge does not concentrate in thin or low-volume pairs, ruling out a staleness artifact, and it replicates on three independent exchanges, on quote-midprice bars, and under alternative return conventions. Same-asset natural experiments (listed spot-Bitcoin ETFs against Bitcoin, listed GLD against OTC spot gold) test whether the intermediation of an instrument's own venue removes the reversal from its prices, and it does not: a heavily intermediated, regular-hours exchange-listed wrapper inherits it at full size, while listed equities merely correlated with Bitcoin do not. The reversal is thus located in the price process formed where primary price discovery happens, and survives transmission into a heavily intermediated wrapper. Its spot-market value is nonetheless below 1.5 basis points per trade, an order of magnitude inside realistic transaction costs, which is consistent with a limits-to-arbitrage account of its persistence.

Keywords: market efficiency; mean reversion; cryptocurrency; market microstructure; high-frequency data; constrained logistic regression; kinetic Ising model

JEL classification: G14; G12; C58; C53

1 Introduction

1.1 The question

Market efficiency is a matter of degree rather than a binary property [14, 17, 29]. Different markets, at different horizons, absorb information at different speeds, and the residual predictability left behind is a measurable signature of their microstructure. This paper measures that signature at the shortest horizons (5 minutes to 4 hours; the 5-minute panel is crypto-only), focusing on cryptocurrency markets with exchange-listed US equities as the disciplining contrast. We ask two coupled questions: how strong is short-horizon *directional* (sign-of-return) predictability in crypto relative to the most heavily intermediated continuously-traded markets, and what shape does it take, trend-following or mean-reverting, and with what memory decay? We answer both at population scale with one interpretable model and one matched protocol applied to 183 crypto pairs and 187 US stocks/ETFs, and, as supporting context, to a focal panel of fifteen instruments across six asset classes. The population-level claim is crypto vs. US equities; the FX, commodity and bond readings come from the focal panel alone and are framed as suggestive.

A cross-market contrast, however strong, cannot by itself say *what the predictability is a property of*, because every such comparison varies the asset and the venue together. Two explanations survive it equally well: that the reversal is a property of each trading venue separately, so that a heavily intermediated venue competes it out of its own prices whatever the underlying does, or that it is a property of the *underlying price process*, formed wherever primary price discovery happens, in which case any instrument priced off that process should carry it however heavily its own venue is intermediated. The two make opposite predictions for a listed ETF on a 24/7 underlying, so we discriminate between them directly (Section 5.5), holding the asset fixed and varying only the structure it trades in. The experiment identifies which venue matters (the instrument's own or the one where the underlying price is formed), not why the underlying reverts, and we state its conclusions in those terms throughout.

The model is a three-parameter autoregressive logit whose lag weights are constrained to a smooth decaying kernel; algebraically it is the Glauber form of a one-dimensional kinetic Ising chain (Section 3), a lineage we use for vocabulary and for the design of the constraint, not as a physical claim about markets. The protocol is a rolling, strictly out-of-sample walk-forward with block-bootstrap inference, false-discovery-rate control, an explicit treatment of cross-sectional dependence and negative controls. The contrast itself does *not* depend on the constraint (Section 5.4); what the constraint buys is calibration and log-loss power in a weak-signal regime (Section 6.1).

1.2 Relation to the literature

Spin models of financial markets have a long econophysics history as models of interacting agents whose binary states aggregate into price moves [4, 5, 21], alongside related herding models built on random-graph rather than lattice interactions [11]. Our use is more literal. We treat the *realized sign sequence* of returns as the spin chain and fit its one-step transition law, following the kinetic-Ising inference methods of Campajola et al. [6] and the equivalence between kinetic Ising dynamics and discrete autoregressive processes established by Campajola et al. [7]; our contribution is a deliberately restrictive parameterization, a Dyson-type long-range kernel [13] on Glauber dynamics [15], used as a *measurement device* uniformly across markets. A second strand measures crypto market efficiency with unconditional scaling statistics. Several studies find Bitcoin's inefficiency real but shrinking as the market matures [2, 12, 38, 39]; others find it episodic rather than trending, with inefficiency persisting throughout and punctuated by efficient spells after bubble deflations [26]. Section 5.4 connects our conditional measurement to that toolkit asset-by-asset. Finally, the microstructure literature on short-horizon reversal in equities [8, 28, 34] and on the structure of crypto markets [32] supplies the interpretive frame for the cross-market contrast we find (Section 8).

1.3 Contributions

1. A matched, out-of-sample population test of short-horizon directional predictability over 183 crypto pairs and 187 stocks/ETFs, under one primary test fixed in advance of the wide-universe run (Section 4), with block-bootstrap inference, false-discovery control, a dependence-aware joint bootstrap and negative controls.
2. The finding that 15-minute directional predictability is concentrated in cryptocurrency markets (90% vs. 3% significant after FDR control; class-mean AUC gap +0.031, 95% CI $[0.027, 0.035]$), mean-reverting in sign, decaying by 4 h, stable across 2021–2026, robust to flat-bar, microstructure-artifact and regular-trading-hours tests, and replicated on three independent exchanges, on quote-midprice bars and under alternative return conventions (Section 6.4), with spot FX intermediate. It is reproduced by parameter-free statistics, a multi-lag sign autocorrelation and the DFA Hurst exponent (Section 5.4), so it is a property of the data, not the estimator.
3. Two same-asset natural experiments discriminating between the venue-based and process-based accounts by holding the underlying fixed: listed spot-Bitcoin ETFs against Bitcoin, listed GLD against OTC spot gold (Section 5.5). A US-listed, regular-hours ETF with designated market makers inherits the reversal at full size while listed equities merely correlated with Bitcoin do not. The wrapper's own intermediation does not remove what the underlying process brings, and correlation without a

mechanical price link transmits nothing. Market structure re-enters only at a second margin, survival past the one-bar boundary, on thinner evidence.

4. A methodological case for the kernel-constrained logit as a low-variance measurement device. The contrast is *estimator-robust*: the unconstrained AR(12) logit reproduces the class-mean gap at $+0.026$ $[+0.023, +0.029]$ and the model-free statistic reproduces it with nothing fitted. What the constraint adds is *calibration*. At AUC $\approx$0.53 the free logit's probabilities are too noisy to beat the base rate on log-loss ($p \approx 0.07$–0.34 even on crypto-15 m) while the three-parameter tie clears it ($p < 10^{-3}$), and it is not beaten by norm-penalized or tuned machine-learning baselines (Section 6.1). The exact kinetic-Ising $\leftrightarrow$ AR-logit equivalence (building on [7]) comes with an identifiability analysis (Appendix A).
5. A cost-of-capture account: thresholded accuracy rises above 56% on the most confident decile, yet the gross edge per trade stays an order of magnitude inside spot transaction-cost bands in all 183 crypto pairs: a quantified *limits-to-arbitrage* interpretation of the inefficiency's persistence (Section 7).

2 Data

We study three groups over a common calendar window where all assets have data, 2025-01-01 to 2026-02-11, at intervals 15 m / 1 h / 4 h (crypto additionally at 5 m). All series trade continuously or near continuously: the US-listed instruments are sourced as 24-hour bars (1–7% of focal-instrument bars are flat, mostly in the thin overnight session; a regular-trading-hours-only robustness check appears in Section 6.2), and the FX series are 24/5 spot quotes with closed-market bars removed. The focal panel is fifteen instruments across six asset classes: crypto BTC/ETH/SOL/XRP (Binance spot klines); equity indices SPX (SPY), QQQ and small-cap IWM, single stocks AAPL/NVDA/TSLA, gold GLD and 20y+ Treasuries TLT (all Alpaca); and spot FX EURUSD/USDJPY/GBPUSD (Dukascopy).

Each candle t carries a signed intra-candle return $r_t = (\mathrm{close}_t - \mathrm{open}_t)/\mathrm{open}_t$ and a direction label $\mathrm{up}_t = \mathbf{1}[r_t > 0]$; predictors at t are the returns of candles $t-1, \dots, t-N$. The convention assigns flat candles ($r_t = 0$) to the "down" class; Section 6.3 quantifies this. Base rates $P(\mathrm{up})$ lie in 48–53% everywhere, so the target is essentially drift-free and any AUC departure from 0.5 reflects *conditional* skill rather than drift.

Four auxiliary datasets extend the focal panel: (i) a deeper within-crypto dataset (BTC from 2024-03, 5 m–4 h) for the regularization study of Section 6.1; (ii) multi-year 15-minute histories back to 2021-01-01 (four coins, eight US-listed instruments) for the regime-stability analysis of Section 6.2; and (iii) the wide 15-minute universe of Section 5.3: the 183 highest-volume Binance USDT pairs and 187 liquid US stocks/ETFs; and (iv) the same-asset natural-experiment instruments of Section 5.5: spot gold (XAUUSD, Dukascopy) and the listed wrappers IBIT, FBTC, COIN and MSTR (Alpaca), each over the same common span. Construction, cleaning, timezone, and survivorship details for every source, including the external-validation data of Section 6.4, are collected in Appendix B.

3 Model: a kernel-constrained AR-logit and its kinetic-Ising reading

The model is a three-parameter constrained distributed-lag logit. The kinetic-Ising equivalence below motivates the kernel shape and supplies vocabulary; it adds no empirical content beyond the constraint. We call the model the *constrained logit*; tables and figures abbreviate the label to "Ising", and the names are interchangeable.

3.1 The spin transform

Asset returns generally, and crypto returns in particular, are heavy-tailed [10, 12], so a single outlier candle can dominate a linear field. We therefore map each return to a *soft spin*,

$$s_t = \tanh(\lambda r_t) \in (-1, +1), \qquad \lambda = 150, \tag{1}$$

a fixed feature map (not a fitted parameter) that preserves the sign and ordering of typical moves while soft-clipping tail events; the decision boundary is insensitive to λ over a wide range (Section 6.2). Because λ is fixed while volatilities differ across assets, cross-class comparisons of coupling *magnitudes* use a volatility-standardized variant,

$$s_t = \tanh\bigl(c\, r_t/\hat{\sigma}\bigr), \qquad c = 0.45, \tag{2}$$

with $\hat{\sigma}$ the training-window return standard deviation (estimated causally, then frozen) and c matched to $\lambda = 150$ on BTC-15 m; every cross-market conclusion is unchanged under this standardization.

3.2 A kernel-constrained AR(N) logit

Let $\sigma_t \in \{-1, +1\}$ be the realized direction and s_t its soft proxy. The working model is an autoregressive logistic regression on the previous N soft spins,

$$P(\sigma_t = +1 \mid s_{t-1}, \ldots, s_{t-N}) = \text{sigmoid}\Big(C + \sum_{k=1}^{N} w_k \, s_{t-k}\Big), \tag{3}$$

with one restriction: the lag weights are *tied* to a smooth, monotonically decaying kernel,

$$w_k = A \, k^{-\alpha}, \tag{4}$$

with lag distance k in candles ($k = 1$ is the most recent prior candle), so the model has three free parameters (C, A, α) no matter how many lags it sees. The amplitude A carries the economics: $A > 0$ means recent moves push the next move the same way (momentum), $A < 0$ means they push it back (mean reversion); α sets how fast the influence of older candles decays. The power-law form of Eq. (4) is a choice of parameterization rather than an empirical finding. Appendix A shows that the data identify the *decay* of the kernel but cannot distinguish power-law from exponential decay at these sample sizes, and every quantity we interpret (the sign and magnitude of A, the out-of-sample skill) is invariant to that choice.

3.3 The kinetic-Ising reading

Equations (3)–(4) are exactly the one-step transition law of a one-dimensional kinetic Ising chain with Dyson-type long-range couplings [13], which is where the parameterization comes from. For the causal local field $h_t = h_0 + \sum_k J_0 k^{-\alpha} s_{t-k}$, Glauber dynamics [15] give $P(\sigma_t = s \mid h_t) = e^{\beta s h_t}/2\cosh(\beta h_t)$, and setting $s = +1$ recovers Eq. (3) identically with $C \equiv 2\beta h_0$ and $A \equiv 2\beta J_0$. The corresponding multivariate statement is established in [7], whose Theorem 1 maps the kinetic Ising model onto a VDAR(1) process *if and only if* every coupling is non-negative, a restriction that excludes the antiferromagnetic regime we report, so the single-chain identity above, which holds for either sign of A, is what our estimates rest on. The inverse temperature β is not separately identifiable and is absorbed into (C, A), which we estimate directly. We retain the magnetic vocabulary purely as labels: $A > 0$ *ferromagnetic* (momentum), $A < 0$ *antiferromagnetic* (mean reversion).

3.4 Regularization, estimation, and baselines

What the lineage buys is a *shape* constraint. L_1/L_2 penalties shrink weight magnitudes but treat lag 1 and lag 48 symmetrically; the kernel tie instead forces the entire weight vector onto the two-dimensional manifold $\{A \, k^{-\alpha}\}$ (monotone, sign-coherent, decaying), so the effective degrees of freedom do not grow with N. In a weak-signal regime this variance reduction is decisive (Section 6.1).

All models are fit by maximum likelihood: α is profiled over a grid, with (C, A) by unpenalized logistic MLE at each candidate and α chosen by log-loss on a chronological validation tail (the last 20% of the training window) before (C, A) are refit on the full window. The L_1/L_2 and tuned machine-learning baselines select their hyperparameters by the identical protocol, so the constrained model and its competitors differ *only* in hypothesis space, never in objective or in information available. Baselines: free/ridge/lasso AR(12) logits; the training base rate; gradient-boosted trees and an MLP on the same twelve lagged returns (Section 6.1); and, as estimator-independent references, the model-free lag-1 return autocorrelation ρ_1 together with the multi-lag *sign* statistics of Section 5.4, which are the parameter-free counterparts matched to the model's hypothesis space.

4 Statistical protocol

The headline claim rests on a single primary test, fixed in design before the wide universe was run (not, however, a registered pre-analysis plan). To make every researcher degree of freedom explicit, and re-runnable from the replication package (Appendix C), we state it in full.

The main test. *Universe:* 183 Binance USDT spot pairs (highest 24-hour quote volume at selection, excluding stablecoin-quote pairs and leveraged tokens) and 187 liquid US stocks/ETFs (curated large caps plus broad and sector ETFs); exact symbol lists frozen in the replication package. *Data:* 15-minute candles, 2025-01-01 to 2026-02-11; label $\mathrm{up}_t = \mathbf{1}[r_t > 0]$ with r_t the intra-candle open-to-close return. *Models:* the kernel-constrained logit of Eqs. (3)–(4) ($N = 12$, $\lambda = 150$) and the unconstrained free AR(12) logit, both by MLE with selection on a chronological validation tail (last 20% of each training window). *Walk-forward:* train 5,760 / test 960 candles, step equal to the test block; non-overlapping out-of-sample blocks concatenated and scored once. *Statistic and inference:* out-of-sample AUC; one-sided moving-block-bootstrap p-value for AUC > 0.5 (block 384 candles, 1,000 resamples) per model; per-asset conjunction $p_\cap = \max(p_{\mathrm{Ising}}, p_{\mathrm{free}})$; Benjamini–Hochberg FDR at $q = 0.05$ jointly across all 370 assets. *Population claim:* class-mean AUC gap $\Delta = \overline{\mathrm{AUC}}_{\mathrm{crypto}} - \overline{\mathrm{AUC}}_{\mathrm{stocks}}$ under a joint moving-block bootstrap on the shared time grid. *Exclusions:* assets with fewer than train + test candles are skipped; no other exclusions. Pegged stablecoins are retained and flagged; flat-bar sensitivity is quantified separately (Section 6.3).

Everything else in the paper (other horizons, kernel families, confidence thresholds, simulations) is supporting analysis around this one test, and none of it feeds back into the test's choices. This includes the process-versus-venue discrimination of Section 5.5: the hypotheses it separates are stated in Section 1.1 because they frame the paper's interpretive question, but the experiment itself was designed and run *after* the primary test, on instruments outside its universe, and is reported as supporting analysis rather than as a pre-specified confirmation.

The same discipline extends to every horizon and instrument outside the box: same calendar span, same window geometry in candles per interval (1 h $\to$ train 2,160 / test 360; 4 h $\to$ 720 / 120), same models and objective, with strictly causal refits at each step and each instrument's non-overlapping test blocks concatenated and scored once.

4.1 Inference

Because adjacent candles' predictions share overlapping feature histories, all per-asset inference uses a moving-block bootstrap [27] on the concatenated OOS series (block $\approx$4 days; 1,000 resamples), giving a 95% CI for AUC [19] and one-sided p-values for "AUC > 0.5" and "log-loss skill over the base rate > 0," computed *independently* for the constrained and free logits, so that "predictable" means *two different estimators agree.* The finest resolvable p at $B = 1{,}000$ is 10^{-3}; zero exceedances is reported as $p < 10^{-3}$. Two scope caveats: the bootstrap holds the walk-forward parameters fixed, so it quantifies sampling variability *conditional on the fitted models*, if anything understating total uncertainty; and the percentile interval is not an exact no-skill null, for which the shuffled-label control of Section 6.5 is the direct complement.

We quantify the first caveat directly. On a 50-asset subset (the focal coins plus evenly spaced Ising-AUC quantiles of each class, 25 per class) we re-run the *entire* procedure, per-fold α profiling and MLE refits of both models included, inside every resample of a moving-block bootstrap of the raw feature–label series ($B = 200$, the same $\approx$4-day blocks). The refit-inclusive intervals are only marginally wider than the conditional ones (median half-width ratio 1.12, IQR 1.02–1.27), and the conclusions get stronger rather than weaker. 21 of 25 crypto assets remain significant under the full procedure, all of them under both models (BTC $[0.523, 0.536]$, ETH $[0.529, 0.542]$, XRP $[0.526, 0.537]$, each $p < 1/B$; the two losses are thin alt-pairs), while the stock side's two conditional significances (about the 1.3 expected by chance at the 5% level) are eliminated outright ($2/25 \to 0/25$). Conditioning on the fitted models is therefore innocuous for the contrast this paper rests on. The primary metric is AUC; log-loss skill is secondary (proper scoring rule [16], noisier); the fitted (A, α) and the model-free ρ_1 are reported throughout.

Multiple testing Searching over many candidate models invites data snooping [41], but our multiplicity runs across *assets* rather than models: the wide universe tests hundreds simultaneously, the setting in which cross-sectional false-discovery control is the relevant discipline [20]. Preferring discovery power to familywise control, we control the false discovery rate with Benjamini–Hochberg [3] at $q = 0.05$, applied jointly across all 370 assets to the conjunction p-value $p_\cap = \max(p_{\mathrm{Ising}}, p_{\mathrm{free}})$, a valid intersection–union p-value for the null "at least one of the two models has no skill."

Cross-sectional dependence Crypto pairs co-move strongly (mean pairwise correlation 0.40 vs. 0.23 for stocks); under equicorrelation the effective number of independent assets is $N_{\mathrm{eff}} \approx 2.5$ for crypto and 4.2 for stocks, so per-asset significance tallies are *descriptive.* The population-level inference that respects this is a joint moving-block bootstrap ($B = 5{,}000$ resamples): time blocks are resampled once on the shared grid and applied to every asset simultaneously, preserving the cross-sectional dependence; the statistic is the class-mean AUC gap Δ (Section 5.3). Repeating the joint resampling at blocks of 192, 384, and 768 15-minute slots ($\approx$2, 4, and 8 days) leaves every CI well to the right of zero, so the gap is insensitive to the bootstrap block length.

5 Results: predictability across markets

We present the focal panel first because its fifteen instruments anchor the asset-class interpretation; the population claim of the paper is the primary 370-asset test of Section 5.3, and readers interested only in the headline evidence can go there directly.

5.1 Fifteen-minute predictability in the focal panel

Figure 1 and Table 1 give the central result. The three cryptocurrencies sit clearly to the right of AUC $= 0.5$ with 95% bootstrap CIs that exclude it; the eight US-listed instruments cluster on 0.5; the three FX pairs sit in between. (SOL lacks matched 15-minute data here and enters only at 1 h; the independent-venue refetch of Section 6.4 fills the gap at 15 m: AUC 0.529, both models $p < 10^{-3}$.)

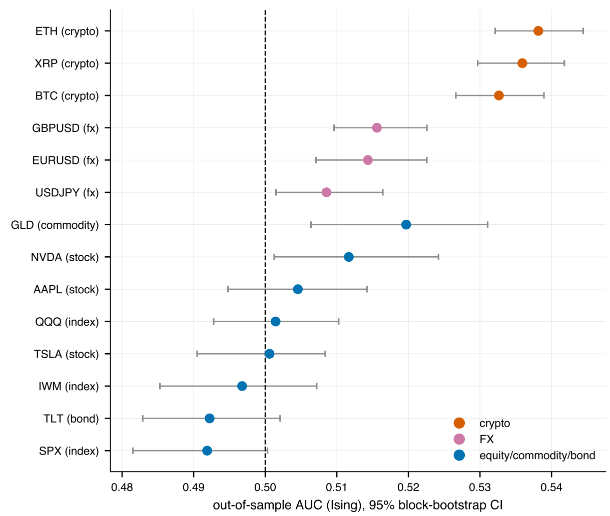

Figure 1: 15-minute out-of-sample AUC per focal instrument with 95% block-bootstrap CIs. Crypto clears 0.5; US-listed instruments cluster on it. *(Central result.)*

Instrument	Class	n_{OOS}	AUC	95% CI	Ising p	Free p	A	ρ_1
ETH	crypto	33,312	0.538	[0.532, 0.544]	$\mathbf{< 10^{-3}}$	$\mathbf{< 10^{-3}}$	−0.31	+0.005
XRP	crypto	33,312	0.536	[0.530, 0.542]	$\mathbf{< 10^{-3}}$	$\mathbf{< 10^{-3}}$	−0.27	−0.023
BTC	crypto	33,312	0.533	[0.527, 0.539]	$\mathbf{< 10^{-3}}$	$\mathbf{< 10^{-3}}$	−0.33	−0.003
GLD	commodity	11,642	0.520	[0.506, 0.531]	**0.005**	$\mathbf{< 10^{-3}}$	−0.10	+0.024
GBPUSD	fx	21,928	0.516	[0.510, 0.523]	$\mathbf{< 10^{-3}}$	**0.002**	−0.74	−0.023
EURUSD	fx	21,930	0.514	[0.507, 0.523]	$\mathbf{< 10^{-3}}$	**0.014**	−0.71	−0.021
NVDA	stock	11,996	0.512	[0.501, 0.524]	0.015	0.258	−0.13	+0.013
USDJPY	fx	21,934	0.509	[0.502, 0.516]	0.008	0.239	−0.13	−0.010
AAPL	stock	11,940	0.505	[0.495, 0.514]	0.176	0.055	−0.13	−0.008
QQQ	index	11,994	0.501	[0.493, 0.510]	0.353	0.116	−0.07	−0.006
TSLA	stock	11,996	0.501	[0.490, 0.508]	0.506	0.211	−0.05	+0.007
IWM	index	11,975	0.497	[0.485, 0.507]	0.744	0.138	−0.05	−0.033
TLT	bond	11,917	0.492	[0.483, 0.502]	0.927	0.643	−0.07	+0.008
SPX	index	12,006	0.492	[0.482, 0.500]	0.972	0.641	−0.05	−0.030

Table 1: 15-minute out-of-sample AUC (Ising), 95% CI, one-sided p for AUC> 0.5 under two models, fitted coupling A (fixed λ; standardized values in Section 6.2), and lag-1 autocorrelation ρ_1. Bold: $p < 0.05$ under both models.

Under the criterion $AUC > 0.5$ *significant under both models*, the tally is crypto 3/3, FX 2/3, US-listed 1/8: strongest in cryptocurrencies (AUC 0.533–0.538); weaker but two-model-corroborated in the two European spot FX pairs (≈0.515) and in gold; nothing in exchange-listed equities, indices, or Treasuries (where ≈0.4 false positives per model are expected among 8 instruments at 5%). The between-group difference is itself significant. Permuting the crypto/non-crypto labels over the per-instrument AUCs (10^5 permutations), the crypto-minus-rest gap of +0.030 (0.536 vs. 0.505) is exceeded by chance with probability $p = 0.003$; split at its midpoint each crypto OOS series holds in both halves (BTC 0.530 / 0.535).

The single-stock couplings are informative: AAPL and NVDA carry a *negative* A (≈ -0.13), the familiar bid–ask-bounce reversal of individual equities [28, 31, 34], yet their AUC barely clears 0.5; crypto's reversal coupling is two-to-three times larger *and* strong enough to rank direction out-of-sample. The model-free ρ_1 on *returns* is ≈0 everywhere, including crypto (BTC −0.003, ETH +0.005); this is not an absence of structure but a cancellation, which Section 5.4 develops as the paper's model-free result.

5.2 Horizon dependence and the coupling

Mean AUC decays with the sampling interval (Figure 2): crypto 0.536 (15 m) → 0.521 (1 h) → 0.496 (4 h); FX 0.513 → 0.507 → 0.487; US-listed instruments ≈0.50 throughout. By the bootstrap, crypto AUC is significantly > 0.5 in 4/4 coins at 1 h under the constrained logit (BTC, ETH $p < 10^{-3}$; SOL $p = 0.008$; XRP $p = 0.012$), with the free logit corroborating 3/4; FX drops to 0/3 two-model significant at 1 h; US-listed is 0/8 at both 1 h and 4 h. This is the profile of a microstructure signal with finite memory: strongest at the highest frequency and gone within hours, with the ordering across market structures holding at every horizon.

The fitted coupling compresses the ordering into one number per class. At 15 m crypto is strongly antiferromagnetic ($\bar{A} \approx -0.30$); indices, single stocks, gold, and bonds sit near zero ($\bar{A} \approx -0.05$ to -0.13, mostly bid–ask bounce); FX is intermediate on the volatility-standardized scale of Eq. (2) ($\bar{A} \approx -0.11$; the fixed-λ values in Table 1 overstate FX magnitudes because FX returns are small relative to $1/\lambda$; Section 6.2). Across horizons, crypto stays negative at 1 h ($\bar{A} \approx -0.13$) before fading toward zero at 4 h, whereas US-listed instruments drift from weak reversal to weak momentum ($\bar{A}$: $-0.08 \to -0.01 \to +0.07$). Short-horizon crypto returns over-react and revert; intermediated markets show no exploitable reversal.

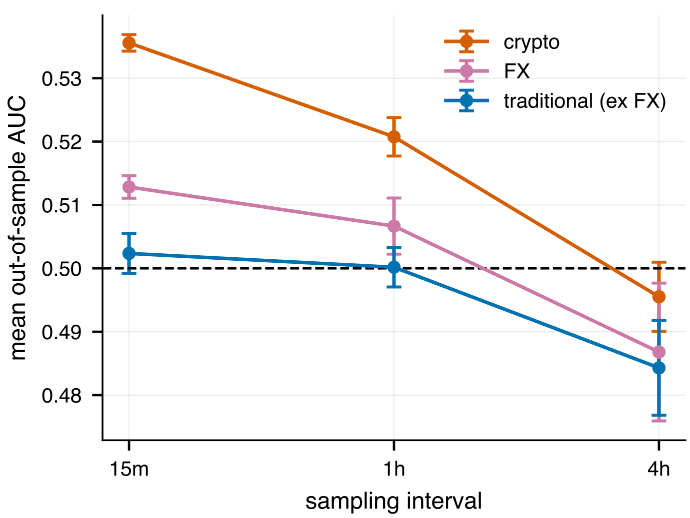

Figure 2: Mean out-of-sample AUC vs. sampling interval (± SE across assets). Crypto's edge is strongest at the highest frequency and decays toward no-skill within hours; US-listed instruments sit at 0.5 throughout, with FX intermediate. The class ordering holds at every horizon.

5.3 A 370-asset universe

To test whether the focal result generalizes, we repeat the matched 15-minute study over a far wider universe of 183 Binance USDT crypto pairs and 187 liquid US stocks/ETFs, each over the common 2025–2026 span with identical walk-forward geometry and per-asset two-model block-bootstrap tests, now under the multiple-testing and dependence framework of Section 4.1. The two classes separate widely (Figure 3).

- **Predictability.** Crypto AUC averages 0.531 (median 0.529; 98% of pairs above 0.5); stocks average 0.499 (median 0.499; 47% above 0.5). After joint Benjamini–Hochberg control at $q = 0.05$ on the conjunction p-values, 90% of the 183 crypto pairs remain significant versus 3% of the 187 stocks. The two AUC distributions barely overlap. (Section 6.3 bounds the contribution of flat-bar label artifacts to these levels: on non-flat bars the class means are 0.520 vs. 0.498, with 97% of crypto pairs still above 0.5.)
- **The dependence-aware population statement.** The joint moving-block bootstrap of Section 4.1 puts the class-mean AUC gap at $\Delta = +0.031$ with 95% CI $[+0.027, +0.035]$ and $p(\Delta \leq 0) < 2 \times 10^{-4}$ ($B = 5{,}000$) under the constrained logit and, independently, $\Delta = +0.026$, CI $[+0.023, +0.029]$ under the free logit. Even at an effective sample of $N_{\text{eff}} \approx 2.5$ dependent "crypto observations," the contrast is unambiguous because it is computed from the time dimension, not the cross-section.
- **Mean reversion.** The fitted coupling is negative in 98% of crypto pairs (mean $A = -0.149$), versus a near-symmetric, centered-on-zero distribution for stocks (mean $A = -0.016$, 57% negative). Antiferromagnetic short-horizon coupling is a near-universal crypto property and essentially absent in equities.

The contrast survives one horizon up Resampling the entire universe to hourly candles and repeating the test, 58% of 169 crypto pairs remain significant after the same FDR control versus 0% of 113 stocks, with crypto AUC attenuating from 0.531 to 0.522 and $A < 0$ still in 96% of crypto pairs. This reproduces the focal horizon decay (Section 5.2) at population scale.

The exceptions fit the pattern The few crypto pairs without significant predictability are dominated by pegged stablecoins and brand-new microcaps; the eleven US-listed instruments clearing the raw two-model test cluster in precious metals and miners (GLD, SLV, GDX, Newmont, Freeport), the most speculative, near-24-hour segment of the listed universe, echoing the lone-gold exception of the focal study.

The gap is not an artifact of ex-post universe selection The crypto universe was frozen on a volume ranking measured *after* the sample (Appendix B). Re-selecting on *ex-ante* (first-sample-month) volume leaves the gap unchanged or slightly larger: the ex-ante top-100 pairs give $\Delta = +0.033$ $[+0.029, +0.037]$ and the 133 pairs already trading ex ante give $+0.035$ $[+0.030, +0.038]$, so late-listed survivors dilute rather than drive the effect. Nor does predictability rise with volume: across ex-ante volume quintiles crypto mean AUC is flat (0.533 to 0.538) and the volume–AUC rank correlation is slightly *negative* (-0.16). Full survivorship (delisted pairs, unavailable from the public API) remains a disclosed limitation.

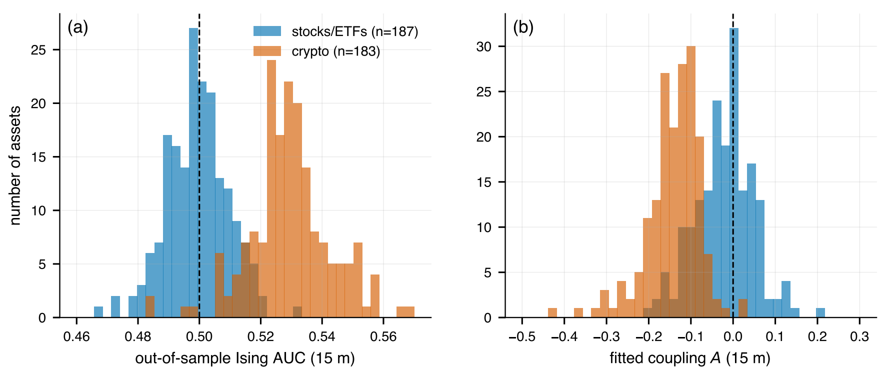

Figure 3: The 15-minute wide universe (183 crypto pairs, 187 stocks/ETFs). (a) OOS Ising AUC: crypto sits to the right of 0.5, stocks centered on it. (b) Fitted coupling A: negative for nearly all crypto pairs, centered on zero for stocks.

5.4 Model-free confirmation: the contrast without any fitted model

The obvious objection to a result at AUC $\approx$0.53 is that it could be a property of the estimator rather than of the data. It is not: the entire cross-market contrast is reproduced by statistics with *no fitted parameters at all*.

The matched statistic is on signs, not returns Contrasting a twelve-lag model against the *one*-lag return autocorrelation ρ_1, the natural first comparison and the one that makes the signal look invisible without the model, is underpowered by construction. The λ-sweep of Section 6.2 shows the effect is unchanged from a near-linear encoding ($\lambda = 50$) to a near-*sign* encoding ($\lambda = 1000$), so the estimator is to a good approximation reading the sign sequence over many lags; its model-free counterpart is the multi-lag autocorrelation of $\sigma_t = \mathrm{sgn}(r_t)$, not the one-lag autocorrelation of r_t. Flat bars enter as $\sigma_t = 0$, so the statistic is flat-bar-artifact-free by construction. Measured that way the structure is not weak at all (Table 2, Figure 4a): across the 183 crypto pairs the mean $\rho_1^{\mathrm{sign}} = -0.029$ with the sign-sequence Ljung–Box $Q(12)$ significant in 97% of pairs, against -0.004 and 13% for the 187 stocks. Defining the kernel-weighted sign correlation

$$R = \mathrm{corr}\Big(\sigma_t,\ \sum_{k=1}^{12} k^{-\alpha}\sigma_{t-k}\Big), \qquad \alpha \equiv 1 \text{ fixed a priori}, \tag{5}$$

(no fitted quantity enters), R is negative in 98% of crypto pairs, exactly the fraction for which the fitted A is negative, versus 50% of stocks. The joint moving-block bootstrap of Section 4.1, applied to R on the shared grid, puts the class gap at -0.035, 95% CI $[-0.041, -0.029]$, $p < 2 \times 10^{-4}$, and across all 370 assets R correlates -0.72 with the model's out-of-sample AUC (Figure 4d). The fitted model is largely an efficient way of scoring a dependence a two-line statistic already exposes.

	ρ_1 (returns)	ρ_1 (signs)	Ljung–Box $Q(12)$	R (Eq. 5)
BTC	-0.003	-0.038	$p \approx 10^{-20}$	-0.052
ETH	$+0.005$	-0.056	$p \approx 10^{-38}$	-0.070
XRP	-0.022	-0.044	$p \approx 10^{-22}$	-0.055
SOL	-0.017	-0.038	$p \approx 10^{-16}$	-0.047
Wide crypto (183)	-0.046	-0.029	sig. in **97%**	-0.035 (98% < 0)
Wide stocks (187)	-0.009	-0.004	sig. in **13%**	-0.000 (50% < 0)

Table 2: The model-free statistic matched to the model. The lag-1 return autocorrelation is ≈0 on the focal coins; the same series carry a strongly significant multi-lag *sign* autocorrelation. R is the kernel-weighted sign correlation of Eq. (5) with α fixed at 1 a priori; no parameter is fitted anywhere in this table.

Why ρ_1 on returns vanishes The cancellation is mechanical, and it also explains the generative shortfall of Section 6.5. Binning each bar by the decile of $|r_{t-1}|$ (flat bars excluded, so illiquid pairs cannot inflate the rate), the class-mean sign-flip rate rises *monotonically* through all ten deciles, from 50.2% at the smallest moves to 53.0% at the largest, and the top-three-decile mean exceeds the bottom-three in 96% of the 182 crypto pairs (the one stablecoin is dropped for degenerate deciles); the stock class mean stays between 49.7% and 51.2% (Figure 4c). On BTC the rate rises from 49.3% in the smallest decile to 53.9% in the largest. But ρ_1 weights each pair by $r_{t-1}r_t$, so it is dominated by the largest decile, whose contribution to ρ_1 is $+0.0036$, opposite in sign to and roughly cancelling the -0.0071 contributed by deciles 4–9. Short-horizon crypto reversal is thus a *sign* phenomenon that strengthens with move size while its magnitude-weighted trace nearly vanishes. That is what the bounded spin $\tanh(\lambda r)$ is for, since it caps the influence of the top decile without discarding it. The same asymmetry explains why a Glauber chain fitted to these data over-produces ρ_1 and variance-ratio signatures (Section 6.5), namely that it propagates the sign structure as though it were magnitude structure.

Two further model-free readings DFA-1 Hurst exponents [22, 33] of the same 15-minute returns are antipersistent for crypto (mean $H = 0.469$, median 0.473; 89% of 183 pairs below 1/2) and not for stocks (mean $H = 0.510$; 28% below 1/2). The crypto reading corroborates Alvarez-Ramirez et al. [1], who report Bitcoin alternating between efficient periods and anti-persistent ones (H below 1/2) at day, hour and second frequencies; we recover the same sign at 15 minutes and, with the 183-pair cross-section, show that it separates crypto from equities. Under the same joint, dependence-preserving block bootstrap the class-mean H gap is -0.049, 95% CI $[-0.084, -0.012]$, $p < 2 \times 10^{-4}$ (the bootstrap re-estimates H on scales capped at half the block length, and the point estimate is recomputed on that same capped scale set). The per-asset H co-varies with the fitted coupling ($r = +0.29$; within crypto $r = +0.21$) and with OOS AUC ($r = -0.44$). Lo–MacKinlay variance ratios [30] sit below unity at every aggregation level for crypto (group means

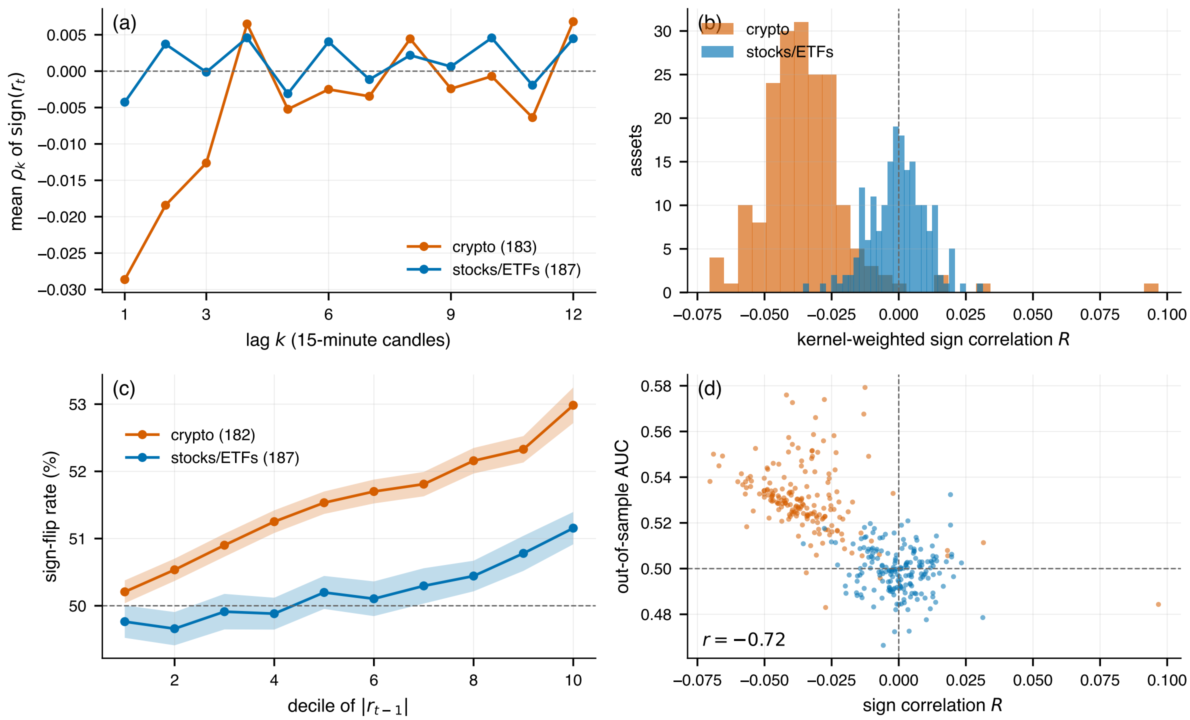

Figure 4: The cross-market contrast without a fitted model. (a) Mean autocorrelation of $\mathrm{sgn}(r_t)$ by lag: crypto shows a decaying reversal kernel over $\sim$3 candles, stocks sit on zero. (b) The kernel-weighted sign correlation R of Eq. (5) ($\alpha \equiv 1$, nothing fitted) separates the two classes as cleanly as the fitted coupling in Figure 3b. (c) Class-mean sign-flip rate $P(\mathrm{sgn}\, r_t = -\,\mathrm{sgn}\, r_{t-1})$ by decile of $|r_{t-1}|$, flat bars excluded; bands are ± 2 s.e. of the cross-asset mean. Crypto reversal strengthens monotonically with the size of the preceding move and exceeds the stock rate at every decile, which is why the magnitude-weighted ρ_1 on returns nearly cancels. (d) R against the fitted model's out-of-sample AUC, one point per asset: the two-line, nothing-fitted statistic predicts the model's skill ($r = -0.72$).

VR(2/4/8/16) $= 0.993/0.963/0.946/0.945$) and below the US-listed average ($0.997/0.984/0.980/0.982$), with FX in the same ordering (GBPUSD VR(16) $= 0.916$, the only individually significant value, $z = -2.2$).

The three differ in power. The sign statistics certify the contrast per asset and at population level; H certifies it at population level only (its within-class correlation with realized AUC is $r = -0.03$); the variance ratios, being second-moment and magnitude-weighted, are individually insignificant almost everywhere, for the reason diagnosed above. What the conditional walk-forward adds is not detection but a per-asset, out-of-sample, calibrated test of a dependence already visible unconditionally.

5.5 Holding the asset fixed: two natural experiments

Every comparison so far varies the asset and the venue *together*: crypto assets on crypto venues against equity assets on equity venues, leaving the two accounts of Section 1.1 (reversal as a property of each venue separately versus of the underlying *price process*) observationally equivalent. They diverge on one prediction: whether an instrument priced off a 24/7 underlying, but traded in a heavily intermediated venue of its own, carries the effect. Two experiments test exactly that (Table 3). *Gold:* spot XAUUSD (Dukascopy, 24/5, OTC, dealer-intermediated, no consolidated tape) against GLD, the listed ETF on the same metal. *Bitcoin:* Binance BTC spot (24/7, no obligated market makers) against IBIT and FBTC, listed US spot-Bitcoin ETFs whose NAV tracks the same price process, and against COIN and MSTR, listed equities merely *correlated* with Bitcoin rather than priced off it.

The wrapper's venue does not remove the effect The regular-trading-hours cell is the one that discriminates. On those bars IBIT and FBTC both score AUC 0.525 ($p = 0.004$ and 0.008), statistically indistinguishable from BTC restricted to the *same* 09:30–16:00 slots (0.524) and more than three standard deviations above the 187-stock RTH distribution of Section 6.2 (mean 0.498, SD 0.0085; IBIT and FBTC exceed all 187). A listed, US-regulated, RTH-traded ETF with designated market makers therefore carries

Instrument	Structure	24 h bars		RTH only	24 h + one-bar gap	
		AUC	A	AUC	AUC	A
XAUUSD	OTC spot 24/5	**0.513**	−0.22	0.503	**0.508**	−0.10
GLD	listed ETF	**0.520**	−0.10	0.507	**0.511**	+0.04
BTC	crypto spot 24/7	**0.533**	−0.33	**0.524**	**0.514**	−0.19
IBIT	listed spot-BTC ETF	**0.525**	−0.24	**0.525**	0.508	−0.16
FBTC	listed spot-BTC ETF	0.514	−0.18	**0.525**	0.504	−0.12
COIN	listed equity proxy	0.505	−0.09	0.520	0.492	−0.02
MSTR	listed levered proxy	0.508	−0.08	0.506	0.499	−0.03

Table 3: Same underlying, different market structure. Identical 15-minute walk-forward, models, and block bootstrap throughout; RTH restricts to 09:30–16:00 New York (*including* BTC and XAUUSD, so the session comparison is like-for-like); the gap column re-runs with one candle skipped between the last predictor and the label. Bold: significant under both models at 5%.

the reversal at essentially full size. The NAV-tracking wrappers are also the only listed instruments here that clear the two-model criterion: COIN and MSTR, with high Bitcoin beta but prices of their own, fail it in every cell, on 24-hour bars (0.505, 0.508) and on RTH bars alike. One qualification: COIN's RTH reading is 0.520 and *is* significant under the constrained logit ($p = 0.001$), failing only because the free logit does not corroborate it ($p = 0.36$), so the separation is clean under the stated criterion but not uniform cell by cell; COIN, a crypto-exchange operator whose revenues track crypto volume, is the least surprising place for that. What is inherited is exposure to the *Bitcoin price process*, not membership in a "crypto-related" category. Between the two accounts of Section 1.1 the test therefore rejects the strong venue account, because intermediation of the instrument's own venue does not compete the reversal out of its prices, which is what that account requires. The reversal is made where the underlying price is formed and is *transmitted* into the wrapper through NAV arbitrage, market-maker hedging against the spot price, and the lead–lag between spot and ETF quotes. These channels are observationally equivalent here and nothing in the design separates them; the experiment does not need to, because its object is where the reversal is made rather than how it travels. What it cannot say is *why* the underlying process reverts. An inherent property of the asset and a product of the primary venue's own structure remain indistinguishable, and Section 8.1 keeps that distinction explicit.

What structure does change is persistence past the one-bar boundary Under a one-candle gap between the last predictor and the label, only the round-the-clock and OTC legs stay significant: BTC retains a multi-lag effect (0.514, $p < 10^{-3}$, $A = -0.19$) and XAUUSD a coupling of -0.10 ($p = 0.016$), while IBIT attenuates to insignificance (0.508, $p = 0.11$) and FBTC to nothing (0.504, $p = 0.54$). The two wrappers do not, however, fail in the same way, so describing both as residual bid–ask bounce [34] would overstate the evidence. IBIT keeps a *reversal*-signed coupling ($A = -0.16$ against BTC's -0.19), so what separates it from BTC is significance rather than direction, and it holds a third of BTC's observations; rescaling BTC's interval to IBIT's sample size still excludes 0.5 (0.514 ± 0.010) while IBIT's does not, but with point estimates that close this is not a difference in kind. GLD's coupling turns weakly *positive* instead ($A = +0.04$, significant under both models): small momentum, not attenuated reversal. We therefore do not build the inheritance conclusion on this cell; it rests on the RTH cell above, where the wrappers plainly do carry the effect.

Gold's reversal lives outside the US session Both gold legs lose two-model significance on RTH bars (0.503, 0.507), consistent with the focal-panel finding that GLD's marginal 24-hour significance disappears in regular hours (Section 6.2) and with the ordering being about *when and where* the tape is thin rather than about the metal.

6 Model adequacy

6.1 The structural prior in a weak-signal regime

Section 5.4 established that *detecting* the contrast does not require this model. What the weak-signal regime (AUC $\approx$0.53) does determine is how much of the signal an estimator can convert into *calibrated* probabilities rather than noise, which is what a deployed decision rule, and the log-loss criterion, actually need. We show this with a within-crypto study on the deeper dataset (BTC/ETH/SOL/XRP, 5 m–4 h, BTC history from 2024-03), fitting every model by the identical MLE objective under the same walk-forward.

- The 3-parameter constrained logit beats the unconstrained AR(12) logit on out-of-sample log-loss in 15/15 cells (Figure 5) (best log-loss 13/15, best AUC 11/15, best Brier 13/15; mean log-loss improvement +0.0017 nats, positive in every cell, largest on the coarse, data-scarce 4 h cells; eth-4h: free AUC 0.489 vs. Ising 0.518). L_1/L_2 shrinkage barely moves the free model: norm penalties do not help a diffuse, ordered signal; the *shape* prior does.
- Coefficient stability: across folds the free AR(24) weight vector flips sign at a given lag in 30% of folds; the constrained kernel is sign-stable in 100%.
- Capacity is a liability. Tuned gradient-boosted trees and an MLP given the same twelve lagged returns under the same walk-forward, with hyperparameters selected per fold on the same validation tail, lose to the three-parameter model on out-of-sample log-loss in all 5 cells (e.g. btc-1h: Ising 0.6918 vs. free logit 0.6935, tuned GBM 0.6933, tuned MLP 0.7005), worst at the data-scarcer 1 h horizon. This mirrors the shallow-beats-deep pattern in low-signal financial prediction, where flexible learners peak at low depth and degrade thereafter given the tiny signal-to-noise ratio [18], and where variance-reducing learners outperform tuning-intensive deep networks on noisy feature spaces [25]; where the predictor set is large relative to the signal, parsimony can win outright [40].

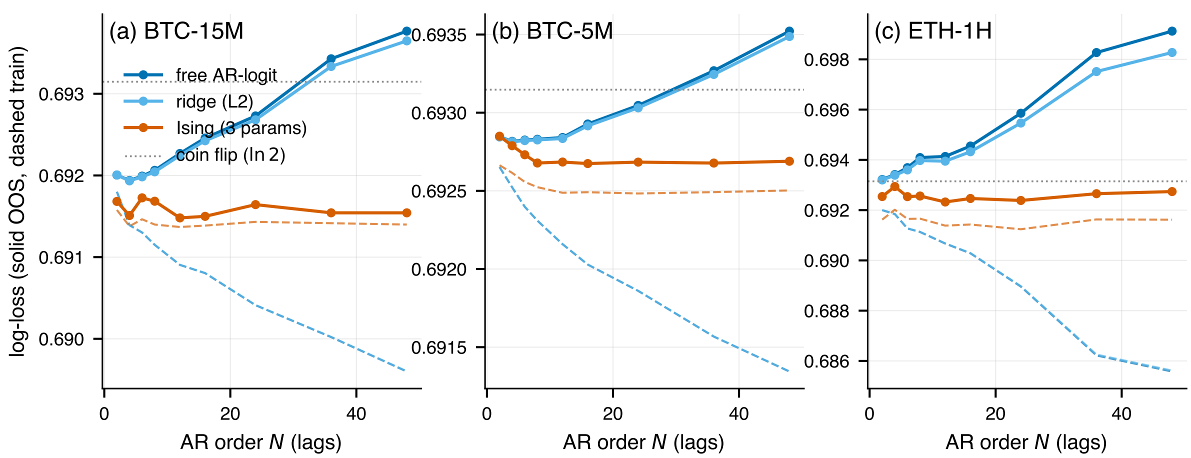

Figure 5: Train (dashed) vs. out-of-sample (solid) log-loss as the AR order N grows, for three within-crypto cells. The free and ridge logits overfit, with the train–test gap widening as capacity grows; the 3-parameter constrained model stays flat with a near-zero generalization gap.

This is why, in the cross-market test, the free logit fails to reach log-loss significance even on crypto-15 m ($p \approx 0.07$–0.34) while the constrained logit clears it ($p < 10^{-3}$). The two estimators *agree on AUC*, which makes the contrast estimator-robust, and *disagree on log-loss significance*, which is what the structural prior buys.

6.2 Robustness battery

We test the headline contrast against five things that could break it: estimator hyperparameters, cross-class scale comparability, trading-session comparability, regime and sample stability, and volatility and session homogeneity. The crypto–equity gap survives all five.

Hyperparameters and standardized spins Sweeping the spin scale $\lambda \in \{50, 100, 150, 300, 1000\}$ (with $N = 12$) and the lag count $N \in \{6, 12, 24, 48\}$ (with $\lambda = 150$), the mean crypto-15 m constrained-logit AUC stays within 0.534–0.536. Refitting every focal instrument with the volatility-standardized transform of Eq. (2) leaves the cross-market picture unchanged: crypto mean AUC 0.535 (fixed-λ: 0.536) with $\bar{A} = -0.32$; non-crypto 0.505 with $\bar{A} = -0.07$. The only material change is cosmetic: the inflated fixed-λ FX couplings of Table 1 (-0.71, -0.74) shrink to -0.14, -0.15, confirming that coupling *magnitudes* should be compared on the standardized scale.

Regular trading hours only A natural objection is that 24-hour stock bars and 24/7 crypto bars are not comparable: a stock's thin, tick-quantized overnight session could either manufacture or mask structure. Restricting the eight US-listed focal instruments to regular trading hours (09:30–16:00 New York, DST-aware) and re-running the matched walk-forward yields mean constrained-logit AUC 0.500 and 0/8 instruments significant under both models (gold's marginal 24-h significance disappears, $p_{\mathrm{Ising}} = 0.23$); the entire 187-stock wide universe on RTH-only bars gives mean Ising AUC 0.498 with only 1.1% of stocks two-model significant, below the 3% on 24-hour bars. Crypto, having no session, is unchanged, so on like-for-like active-session bars the contrast is if anything slightly wider.

Multi-year stability Extending the four focal coins to 2021-01-01 and scoring the same walk-forward per calendar year: the 15-minute edge is significant under *both* models in all 24 coin-years, AUC between 0.523 and 0.550, coupling negative in every cell (Figure 6a). BTC drifts mildly downward (0.544 in 2021 to 0.529 in 2025), consistent with the adaptive-markets literature [23, 26, 29], but five years have not removed the effect. The same holds with nothing fitted: computing the sign correlation R of Eq. (5) per calendar quarter, the crypto class mean is negative in all 21 quarters (range -0.079 to -0.032) and 83 of 84 individual coin-quarters are negative, while the US-listed mean never leaves $[-0.020, +0.007]$ (Figure 6b). The equity counterpart, 40 US-listed instrument-years, yields mean AUC 0.501 (range 0.489–0.513) with 5 isolated two-model significances (no instrument repeats in more than two of its five years; $\approx$2 expected by chance) and couplings near zero ($\bar{A} = -0.04$): five years of equity data, including the 2022 bear market, produce nothing resembling the crypto pattern, so the asymmetry is not an artifact of the 14-month window. This is also where the account is exposed going forward. If walk-forward AUC on future data converges to 0.5, the effect was regime-bound and the adaptive-markets reading wins.

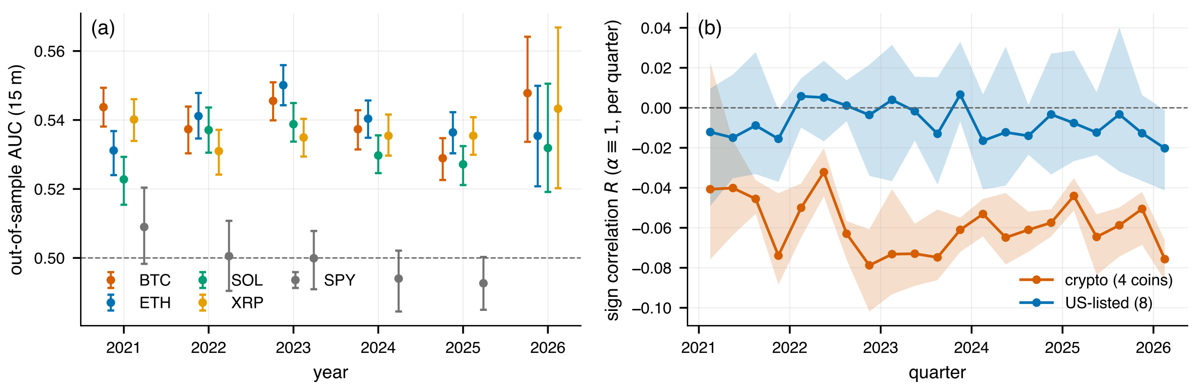

Figure 6: Five years of the same effect, read two ways. (a) Per-year out-of-sample AUC of the constrained model, four focal coins, 2021–2026 (95% bootstrap CIs; 2026 is a partial year): all 24 coin-years sit above 0.5, while SPY (gray), scored identically over 2021–2025, hugs the no-skill line throughout. (b) The model-free sign correlation R of Eq. (5) per calendar quarter (class mean; shading spans the min–max across assets): the crypto mean is negative in every quarter, the US-listed mean stays near zero.

Volatility and session homogeneity Tagging each crypto-15 m out-of-sample prediction with its trailing one-day realized volatility and UTC hour, the AUC is flat across volatility terciles (0.536 / 0.539 / 0.534) and nearly flat across sessions (0.531–0.542). The effect is a standing feature of the tape rather than an episodic or session-specific one.

6.3 Microstructure artifacts: stale prices, flat bars, and the liquidity gradient

The leading mechanical explanation for the headline signature is also the oldest. With last-trade candles, a close recorded at the bid or the ask, or simply stale, mechanically produces one-bar reversal [34], which is exactly a negative coupling, strongest at the shortest horizon. We run four checks: a one-bar gap, a flat-bar exclusion, a liquidity stratification, and a joint microstructure cross-section.

Skipping the most recent bar Genuine multi-lag mean reversion should survive a one-candle gap between the last predictor and the label; pure bounce and staleness, which live at the boundary between adjacent bars, should not. Re-running the identical walk-forward with features lagged one extra candle (predicting candle t from $t-2, \ldots, t-13$), the focal coins retain most of the effect: BTC 0.514, ETH 0.519, XRP 0.513 (all $p < 10^{-3}$ under *both* models), couplings still −0.12 to −0.19. At population scale crypto mean AUC moves from 0.531 to 0.522 (71% of the excess survives), with 81% of the 183 pairs still two-model significant and the coupling negative in 96%; stocks remain at 0.499 either way (Figure 7a). The signal is not a one-bar boundary artifact.

Flat bars (primary robustness table) The convention up $= \mathbf{1}[r > 0]$ assigns flat candles to the "down" class, and flat 15-minute bars are common in thin or tick-quantized instruments (median 16% across the wide US-listed universe, 6% across wide crypto, < 1% in the focal coins); a reversal model scores such bars "correct" while earning exactly zero. This is the most consequential label artifact, so the flat-bar-robust result is a primary table (Table 4): excluding flat bars, wide-crypto mean AUC falls from 0.531 to 0.520 with 97% of crypto pairs still above 0.5, stocks move 0.499 to 0.498, and the focal coins are unaffected (BTC 0.5327 → 0.5326). Accuracy is far more sensitive: the US-listed instruments' apparent 57.9% on high-confidence candles collapses to 50.1% on non-flat bars. This is why the headline metric is flat-bar-robust AUC corroborated by signed-return edge, never raw accuracy.

	Mean OOS AUC		Frac. AUC > 0.5	
Group	All bars	Non-flat	All bars	Non-flat
Wide crypto (183)	0.531	**0.520**	0.98	**0.97**
Wide stocks (187)	0.499	0.498	0.47	0.46
Focal coins (BTC)	0.533	0.533	—	—

Table 4: Flat-bar-robust predictability. Recomputing each asset's out-of-sample AUC on non-flat bars only (dropping $r_t = 0$ candles, which the up $= \mathbf{1}[r > 0]$ convention assigns to "down") leaves the crypto–stock gap intact. This non-flat AUC, not the all-bar value, is the metric we treat as primary.

The liquidity gradient and a microstructure cross-section A thinness artifact would concentrate in the least liquid pairs. Stratifying the 183 pairs by average daily quote volume, raw AUC does decline modestly with liquidity (quintile means 0.539 to 0.527; Spearman $\rho = -0.23$, $p = 0.002$), but the gradient is carried almost entirely by the flat-bar effect: on non-flat bars the quintile means are 0.519, 0.519, 0.518, 0.522 and 0.524, flat across the spectrum (Figure 7b), with the most liquid quintile keeping 86% of pairs two-model significant. Entering the covariates jointly, we regress each pair's flat-bar-robust AUC on standardized log dollar volume, flat-bar fraction, and median $|r_t|$, with HC1-robust standard errors ($R^2 = 0.18$). The intercept is 0.520 ($t = 737$), so after removing the flat-bar channel and conditioning on volume and volatility, the average crypto pair still sits two percentage points above no-skill. The volume slope is *positive* and insignificant (+0.0015, $t = 1.6$); predictability does not rise as pairs get thinner, the opposite of what a staleness artifact requires. Only the flat-bar fraction carries a significant loading (−0.0037, $t = -2.4$). The reversal is a property of the active crypto tape, not of illiquid microcaps; the depth-, imbalance- and fee-resolved tests that would pin the mechanism require order-book data we do not have at universe scale (Section 8.1).

6.4 External validity: independent venues, quote-mid bars, and return conventions

The tests above all reuse the same venue, the same last-trade candles, and the same return convention as the headline result, which leaves three single points of failure. This section removes all three with external data.

Independent exchange replication We refetch the four focal coins at 15 minutes over the identical span from three venues with independent matching engines, order flow, and fee schedules: Coinbase (USD-quoted, US-regulated), OKX, and Bybit (USDT). We then rerun the identical walk-forward, models, and bootstrap (Kraken's public API cannot rebuild the span; Appendix B). All 12 venue–coin cells are significant under both models ($p < 10^{-3}$ everywhere), with per-coin AUC within ±0.002 of the Binance value and the coupling mean-reverting in every cell; a four-venue median-close composite reproduces the effect as well. The effect belongs to the common crypto price process rather than to any one venue's prints. (The refetch also backfills the SOL 15-minute series missing from the focal panel: AUC 0.528–0.529, two-model significant on all four venues.)

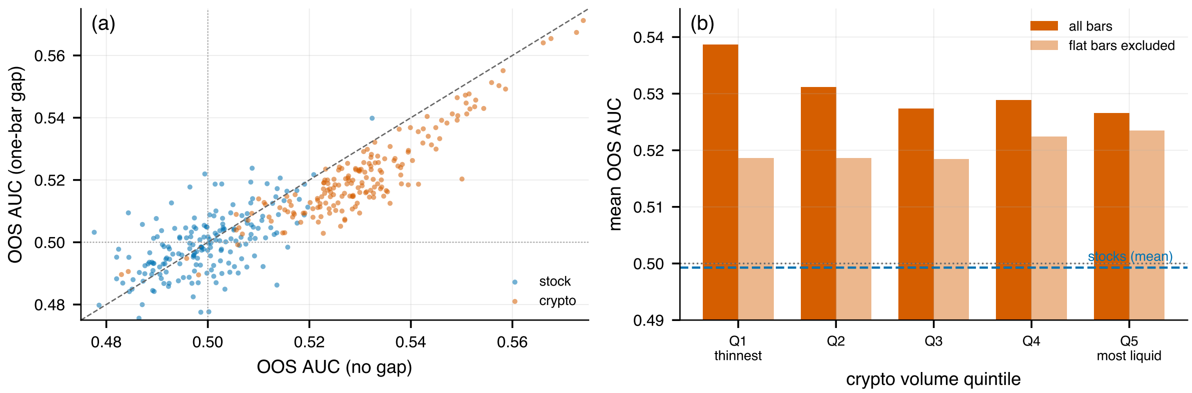

Figure 7: Microstructure-artifact tests on the wide 15-minute universe. (a) Per-asset OOS AUC with vs. without a one-candle gap between features and label: crypto shifts down but stays separated from 0.5 and from stocks. (b) Crypto AUC by volume quintile, all bars vs. non-flat bars: the thin-pair advantage is a flat-bar artifact; the artifact-free signal is uniform. Dashed line: stock mean.

Quote-midprice bars Dukascopy distributes separate bid-side and ask-side 15-minute candles for BTC/USD and ETH/USD (the same source as the FX panel); averaging the sides bar-by-bar gives *quote-midprice* bars in which a last-trade print bouncing between bid and ask cannot exist. The effect is unchanged: BTC mid-bar AUC 0.529 $[0.523, 0.536]$ (Binance trade bars: 0.533), ETH 0.537 $[0.531, 0.543]$ (0.538), both models $p < 10^{-3}$, couplings ≈ -0.30, identical under intra-bar and close-to-close mid returns. Bid–ask bounce [34] is thereby excluded on quote data directly. The same data also price the friction, since the median quoted spread here is 5.4 bp of mid for BTC and 17 bp for ETH, versus a thresholded gross edge of ∼1 bp per trade (Section 7).

Return conventions Finally, we rebuild the Binance label and feature series under alternative return definitions (*close-to-close* and *VWAP-to-VWAP*, alongside the baseline *open-to-close* and the *mid-to-mid* variant above) and rerun everything. Open-to-close and close-to-close coincide to the third decimal on every coin (BTC 0.533, ETH 0.538, SOL 0.529, XRP 0.536 under both), since in a continuously traded market the open of bar t *is* the close of bar $t - 1$ up to one print. (The VWAP variant scores higher, AUC 0.564–0.582, but only because intra-bar time-averaging mechanically induces positive serial dependence [42], so it measures predictability of a smoothed index rather than tradable structure.) The headline does not depend on the candle convention.

6.5 Negative controls

A pipeline of this size could in principle manufacture skill on its own, so we check that it does not. We run the identical 15-minute protocol, unchanged, on two controlled transformations of every focal instrument (Table 5).

Shuffled labels Permuting the label series while leaving features untouched destroys all feature–label dependence; any residual "skill" would indict the scoring and inference machinery itself. There is none: mean AUC 0.500 and 0 of 14 instruments significant under the two-model criterion. Because the population claim rests on the wide universe rather than the focal panel, the same control is also run at full scale: shuffling the labels of all 370 wide-universe series and re-running the identical walk-forward, block bootstrap, and joint Benjamini–Hochberg step yields mean AUC 0.499 (crypto) and 0.496 (stocks), a class-mean gap of +0.002 (against the observed +0.031), 3 raw two-model exceedances at the 5% level (≈ chance), and 0 of 370 significant after FDR. The population-scale machinery, FDR step included, cannot manufacture either the per-asset significances or the class gap.

Phase-randomized surrogates IAAFT surrogates [35, 37] preserve each series' marginal distribution and linear autocovariance while destroying nonlinear temporal dependence; labels are recomputed from the surrogate returns, so a look-ahead or alignment bug anywhere in the pipeline would survive the transformation and push AUC far above the observed levels; none does. The crypto signal collapses (BTC 0.533 → 0.506, ETH 0.538 → 0.499, XRP 0.536 → 0.518; 79% of the class-mean excess removed); what survives is the linear component the surrogate preserves by construction. The crypto headline signal is therefore predominantly *nonlinear*, consistent with the near-zero ρ_1 and variance ratios of Section 5.4. A generative check sharpens the point. A pure Glauber chain simulated from the fitted (C, A, α) reproduces the predictive level (simulated AUC 0.535 vs. empirical 0.538 on average) but over-produces linear signatures (ρ_1 of -0.07 to -0.01, $\mathrm{VR}(16) \approx 0.84$, vs. $\rho_1 \approx 0$, $\mathrm{VR}(16) \approx 0.97$ in the data). In the real market the reversal is organized so that its linear trace nearly cancels, which makes the model an adequate predictive reduction but an incomplete generative one. Section 5.4 identifies the specific misspecification, namely that the chain propagates sign structure as though it were magnitude structure, over-producing exactly the linear signatures the data lack.

IAAFT is anti-conservative for session-heteroskedastic series One number in Table 5 needs explanation: the surrogates clear the two-model criterion on 5 of 11 non-crypto instruments while the *real* series clear it on 3, against ≈0.5 expected; taken at face value, this would say the pipeline manufactures significance from a null. Two explanations fail: it is not a variance effect (mean bootstrap CI half-width 0.0091 on surrogates versus 0.0094 on real data), nor the flat-bar atom being scattered by the rank remapping ($r = -0.07$ between flat-bar share and AUC lift). What explains it is that IAAFT preserves the power spectrum (hence ρ_1) exactly while destroying the volatility clustering and session structure that make that ρ_1 *non-exploitable* in the real series. SPX illustrates: $\rho_1 = -0.030$, among the most negative in the panel, yet its real AUC is 0.492, *below* no-skill, because the linear reversal sits in the thin overnight regime; homogenize the variance and it lifts to 0.515. Across the non-crypto panel surrogate AUC is nearly a pure read-out of ρ_1 ($r = -0.69$) while real AUC is not ($r = +0.31$), and mean intraday heteroskedasticity is 17.6× for listed instruments against 5.0× for crypto. That bias works against us: the row is *biased toward* finding skill on session-heteroskedastic series, so it bounds the linear channel from above and the crypto collapse ($0.536 \to 0.508$) is obtained under an anti-conservative null; its significance counts are not comparable across series with different session structure.

Series	Mean Ising AUC		Two-model significant	
	Crypto	Non-crypto	Crypto	Non-crypto
Real data (Table 1)	0.536	0.505	3/3	3/11
Shuffled labels	0.497	0.501	0/3	0/11
Shuffled labels, wide universe	0.499	0.496	0/370 after FDR	
IAAFT surrogate	0.508	0.505	2/3	5/11

Table 5: Negative controls: the full 15-minute pipeline applied to transformed series (focal instruments; two-model criterion at 5%; the wide-universe row applies the identical shuffle and joint Benjamini–Hochberg step to all 370 assets of Section 5.3). Shuffled labels are a strict null. IAAFT surrogates preserve linear autocovariance while destroying volatility clustering, which makes them *anti-conservative* for session-heteroskedastic series: their non-crypto count (5/11, above the real data's 3/11) bounds the linear channel from above rather than indicting the pipeline, as diagnosed in the text.

7 Economic significance: the cost of capture

This section converts the statistical edge into economic units: from AUC to edge per trade, priced against transaction-cost bands. It adds no statistical evidence beyond Section 5. A deployed strategy acts only when model confidence clears a threshold, $|p_t - 0.5| \geq \tau$, with τ fixed ex ante (we sweep a fixed grid; nothing is selected on the out-of-sample series). The resulting coverage–accuracy–edge curves say two things.

First, the thresholded signal is large in probability units: BTC-15 m directional accuracy rises monotonically from 52.3% (all candles) to 54.6% at $\tau = 0.02$ (38% of candles traded) and 56.2% at $\tau = 0.05$ (9%), and the rise is monotone across confidence deciles, more steeply for the constrained model (49.9% to 55.9% on BTC) than for the free logit (50.8% to 54.7%). Model confidence therefore *ranks* accuracy, though that alone does not show the probabilities are calibrated in level. (Accuracy comparisons are restricted to the focal coins because flat bars inflate accuracy on thin instruments; Section 6.3.)

Second, the same edge is small in return units, and thresholding cannot rescue it: the mean gross edge per trade, $\mathbb{E}[\mathrm{sign}(p_t - \frac{1}{2})\, r_t \mid \text{traded}]$, rises from 0.24 bp of notional on BTC-15 m to ≈1.3 bp at $\tau = 0.05$: significantly positive throughout, yet an order of magnitude inside the cheapest realistic spot round-trip cost band (≈5 bp maker, 10–20 bp taker). Not one of the 183 crypto pairs clears even the 5 bp band at any threshold (the median crypto pair earns 0.46 bp per trade at $\tau = 0.02$; the median stock earns zero), and 5 m is worse rather than better (0.15 bp at $\tau = 0.02$), because higher frequency multiplies opportunities but shrinks each one faster than costs shrink.

The gap between the two is the economically relevant part of the result. A reversal worth half a basis point per candle offers no profit to anyone paying spot fees. Section 8.1 reads its persistence through limits to arbitrage.

8 Discussion

8.1 Process or venue? What the predictability is a property of

The evidence supports a specific answer to the question posed in Section 1.1: short-horizon reversal is a property of the price process of assets whose primary price discovery happens in thin, round-the-clock, lightly intermediated venues, and it is *inherited* by instruments priced off that process even when those instruments trade in a heavily intermediated venue of their own. Venue structure is not irrelevant, since it governs how much survives past the one-bar boundary, but it does not determine whether the underlying reverts.

The cross-market ordering alone cannot establish this, because it is equally consistent with the competing venue-based account. Crypto trades 24/7 across many venues with no consolidated tape, obligated market makers or volatility halts, and is segmented enough that large arbitrage spreads persist for days [32]; its order flow is also far more retail-driven [24], a plausible source of the overreaction a reversal coupling corrects. Spot FX, decentralized and 24/5 but dealer-intermediated, shows a faint copy (AUC $\approx$0.515, standardized $\bar{A} \approx -0.11$) decaying by 1 h, while exchange-listed instruments show nothing, consistent with professional market makers competing short-horizon reversal away well inside the bar, estimated at longer than five minutes but largely complete within thirty for actively traded NYSE stocks [8], and faster in today's electronic markets [9]. But every one of these comparisons varies the asset and the venue together, so the ordering is a signature both accounts predict.

The natural experiments of Section 5.5 separate them, and were built so that either answer was available: the venue account predicts a US-listed ETF with designated market makers, trading only in regular hours, should show no reversal even on a 24/7 underlying; the process account predicts it should show the underlying's. The data reject the strong venue account: the listed wrappers carry the underlying's reversal at full size while COIN and MSTR, correlated with Bitcoin but not mechanically linked to it, carry none (Section 5.5). The positive claim this supports is deliberately narrower than "process, not structure": the reversal is generated where primary price discovery happens (itself a thin, round-the-clock, lightly intermediated environment), and the intermediation of a downstream venue neither removes it nor is needed to explain it.

Venue structure re-enters at a second margin, where the one-bar-gap control does the separating: BTC retains a significant multi-lag effect under the gap ($A = -0.19$, $p < 10^{-3}$) while its listed wrappers do not, and the gold legs diverge in kind: XAUUSD keeps a reversal coupling ($A = -0.10$) while GLD's turns weakly *positive* ($A = +0.04$). Intermediation therefore appears to shape what survives past the bar boundary and in what direction, rather than whether the underlying reverts at all, though the wrappers differing in sign is a caution that this margin rests on a far thinner base than the inheritance result. Consistently, the wide-universe exceptions are gold, silver and miners, listed claims on near-24-hour underlyings, and the horizon decay persists: by 4 h crypto reaches the no-skill line where intermediated markets sit throughout.

The cost-of-capture analysis adds to this from the economic side. In every market we measure, what survives is a residual priced *below* the round-trip cost of its marginal arbitrageur [17, 36], large enough to measure but too small to remove. That is the pattern a limits-to-arbitrage account predicts, though other mechanisms are not excluded. The evidence remains observational, and the natural experiments identify the asset-versus-venue margin rather than the mechanism behind the reversal itself. Pinning that mechanism is the principal open test. The variables that would resolve it (quoted spread and depth, trade count, order-flow imbalance, fragmentation, fee variation) require order-book data we do not have at universe scale, and a panel in which the coupling A failed to co-move with these intensities would weaken the limits-to-arbitrage reading.

8.2 Limitations and open tests

- **Cross-sectional dependence bounds population claims.** With $N_{\text{eff}} \approx 2.5$ effectively independent crypto observations, "90% of 183 pairs" describes breadth, not 183 confirmations; the joint bootstrap gap CI (Section 5.3) is the defensible statement.
- **The wide crypto universe is a survivor set, and single-exchange.** Frozen on a post-sample volume ranking; the ex-ante re-selection of Section 5.3 rules out the volume-conditioning and late-listing components, but a residual survivorship caveat stands (Appendix B) and the 183-pair cross-section remains Binance-only (the focal coins replicate across venues and on quote-midprice bars).
- **Coverage.** The matched window is 14 months; FX (three pairs) and the wide universe stay confined to the 2025–2026 regime. At 4 h, $n_{\text{OOS}} \approx 10^3$ with wide CIs, so the apparent crypto-reversal / equity-momentum flip there is suggestive only. A broad FX panel that failed to sit between crypto and equities would break the market-structure ordering.
- **The natural experiments rest on few instruments**: two spot-BTC ETFs and one gold pair over a single 14-month window. Their two halves are not equally secure. That the listed wrappers *do* carry the effect is robust (IBIT and FBTC clear the two-model test on RTH bars and exceed all 187 stocks in the RTH universe) and is what rejects the strong venue account. The second-margin claim, that intermediation governs survival past the one-bar boundary, is weaker: BTC 0.514 vs IBIT 0.508 is a difference in significance more than in kind, and the wrappers diverge in sign under the gap test (IBIT $A = -0.16$, GLD $A = +0.04$). The decisive extension is a wider wrapper panel, listed ETFs on other thin, round-the-clock underlyings, where a failure to inherit the underlying's coupling would overturn the process-inheritance account. A first version of such a panel, run after the present analysis was frozen, is consistent with inheritance.
- **The economic analysis is gross.** The selective-participation edges ignore venue depth, latency, and adverse selection.
- **The model is a predictive reduction.** The generative check (Section 6.5) shows the fitted chain over-produces linear mean-reversion signatures; it is not a data-generating process for the joint sign–magnitude dynamics.

9 Conclusions

Using a three-parameter, kernel-constrained autoregressive logit (algebraically the Glauber form of a kinetic Ising chain) as a low-variance measurement device, we mapped short-horizon directional predictability under one matched, significance-tested protocol, from fifteen focal instruments to 183 crypto pairs and 187 stocks/ETFs. First, 15-minute directional predictability is concentrated in cryptocurrency markets (90% vs. 3% significant after FDR control; class-mean AUC gap +0.031, 95% CI $[0.027, 0.035]$), mean-reverting, decaying within hours, stable across 2021–2026, replicated on independent venues, quote-midprice bars and alternative return conventions, robust to flat-bar, regular-trading-hours and negative-control tests, with spot FX intermediate. Second, it is not an artifact of the estimator: the unconstrained AR(12) logit reproduces the class-mean gap at +0.026 and a parameter-free multi-lag *sign* statistic reproduces it at -0.035 $[-0.041, -0.029]$, as does the DFA Hurst exponent. What the kernel constraint contributes is calibration rather than detection, which is what makes its probabilities usable and not just its rankings. Third, holding the asset fixed while varying the venue rejects the venue-local account: the reversal is not eliminated by placing the exposure in a listed, market-maker-intermediated wrapper (Section 5.5); it is a property of the price process formed in the primary market, transmitted to instruments priced off it, with intermediation governing only survival past the one-bar boundary. Fourth, the per-trade edge stays an order of magnitude inside spot transaction costs in all 183 crypto pairs, consistent with a limits-to-arbitrage reading of its persistence.

Declaration of competing interest

The author operates an automated trading system for cryptocurrency binary markets that uses a variant of the model studied here. No data from that system are used in this paper. No other competing interests.

Data availability

All market data are obtainable from public sources (Binance, Coinbase, OKX, Bybit, and Dukascopy public APIs; Alpaca market-data API); a complete replication package is publicly available at https://github.com/nadav2/short-horizon-reversion (Appendix C).

A Kernel-shape identifiability

Is the *power-law* form of the kernel a finding or a choice? We refit the model with three tied-kernel families under the identical protocol, in all 15 within-crypto cells and all focal 15 m markets: power-law $w_k = A\,k^{-\alpha}$, exponential $w_k = A\,e^{-(k-1)/\tau}$ (both 3 free parameters), and flat $w_k = A$ (2 parameters). The decay is identified: the flat kernel is worse than the power-law kernel in 13/15 cells, significantly so in 9 (paired block bootstrap), and at 15 m the deficit is large (e.g. ETH 0.527 vs 0.544 AUC, $p < 10^{-2}$). The functional form is not: power-law and exponential decay are statistically indistinguishable: the paired AUC difference is below $|0.003|$ in 14/15 cells, its 95% CI covers zero in 8, and its sign flips across cells. We therefore interpret only what is identified: the *existence, sign, and rough decay* of the memory kernel. The power-law parameterization is retained for its lineage (Dyson's long-range 1D model [13]) and because every conclusion is invariant to swapping it for exponential decay; the fitted $\alpha \approx 1$–1.5 should be read as "slow decay over $\sim$3–6 candles," not as evidence for scale-free memory.

B Data construction, cleaning, and universe details

The fetch scripts in the replication package implement exactly what is described here.

Conventions common to all sources All bars are stamped in UTC with bucket-start timestamps on a shared 15-minute grid, so cross-source alignment is exact integer matching. Missing buckets are left absent, never zero-filled or interpolated; gap rates are negligible where this matters (<0.1% of focal-coin grid slots on every venue). The baseline label is the *intra-bar* open-to-close return, so weekend gaps, splits, and dividends never enter a label or feature across a bar boundary.

Crypto (Binance) Spot klines from the public REST API (`/api/v3/klines`). Binance emits a bucket for every grid slot including zero-volume ones, so flat bars are genuine no-trade or no-net-move intervals, not fabricated fills (Section 6.3). The four focal pairs are continuously listed over the full 2021–2026 history used in Section 6.2.

Wide crypto universe: selection date and survivorship The 183-pair universe was selected on 2026-06-08, *after* the end of the sample span, as the highest 24-hour quote-volume Binance USDT spot pairs on that date, excluding pegged-base pairs and leveraged tokens, and requiring ≥5,000 bars within the span. This has two consequences. First, pairs delisted before the selection date cannot appear: the crypto cross-section is a *survivor* universe (the stock list is curated at the same date, so the cross-class *contrast* is symmetric in this respect). Second, selecting on volume measured after the span conditions on ex-post activity; Section 5.3 confirms empirically that this does not drive the gap. Pegged tokens passing the volume screen are retained and flagged: they surface as the no-signal exceptions in Section 5.3.

US-listed instruments (Alpaca) Fifteen-minute bars from the Alpaca market-data API over the full 24-hour session as distributed; prices are unadjusted (labels are intra-bar, so corporate actions affect no label). The RTH rerun restricts to 09:30–16:00 America/New_York. The median flat-bar share is 16% across the wide stock universe (1–7% focal), concentrated overnight; this motivates the flat-bar analysis and the RTH rerun.

Spot FX (Dukascopy) Bid-side 15-minute candles from Dukascopy's public historical feed; closed-market bars (`high == low`, weekends and holiday gaps only) are removed. The same source supplies the bid- and ask-side series behind the quote-midprice bars of Section 6.4.

Natural-experiment instruments The same-asset experiments of Section 5.5 add two sources, both already used elsewhere. *Spot gold* (XAUUSD) comes from the same Dukascopy feed as the FX panel, bid-side 15-minute candles with closed-market bars (`high == low`) removed by the identical rule; 26,022 bars over the span, 0.1% flat. *Listed wrappers* (IBIT, FBTC, COIN, MSTR) come from Alpaca on exactly the terms used for the other US-listed instruments (full 24-hour session, unadjusted prices, intra-bar labels), with flat-bar shares 2.6%, 11.0%, 3.0% and 1.8% respectively; the flat-bar-robust AUC is reported alongside the raw value in every cell, and FBTC's higher share is why its 24-hour reading is weaker than its RTH reading. IBIT and FBTC launched in January 2024 and so cover the full span; no single-coin ETF on the other focal coins has a long enough history to enter.

Independent venues (Coinbase, OKX, Bybit) Public candle endpoints, fetched with retries and deduplicated on the grid; missing-slot rates are ≈0.06% on Coinbase and effectively zero on OKX and Bybit. Kraken is absent because its public OHLC endpoint returns only the most recent 720 bars.

C Reproducibility

Every number, table, and figure in this paper is produced by a pipeline of small Python modules, distributed as a public replication package at `https://github.com/nadav2/short-horizon-reversion`: the pipeline code, the exact symbol lists, fetch scripts that rebuild every input dataset from public sources, the frozen result files with SHA-256 hashes, and a README mapping each table and figure to a single `python -m paper.<module>` command. The environment is Python ≥3.11 under `uv`; all bootstrap and permutation procedures are seeded, so reruns are deterministic given identical inputs. Because the volume ranking is time-dependent, the *frozen symbol lists*, not the rules, define the universe for replication.

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